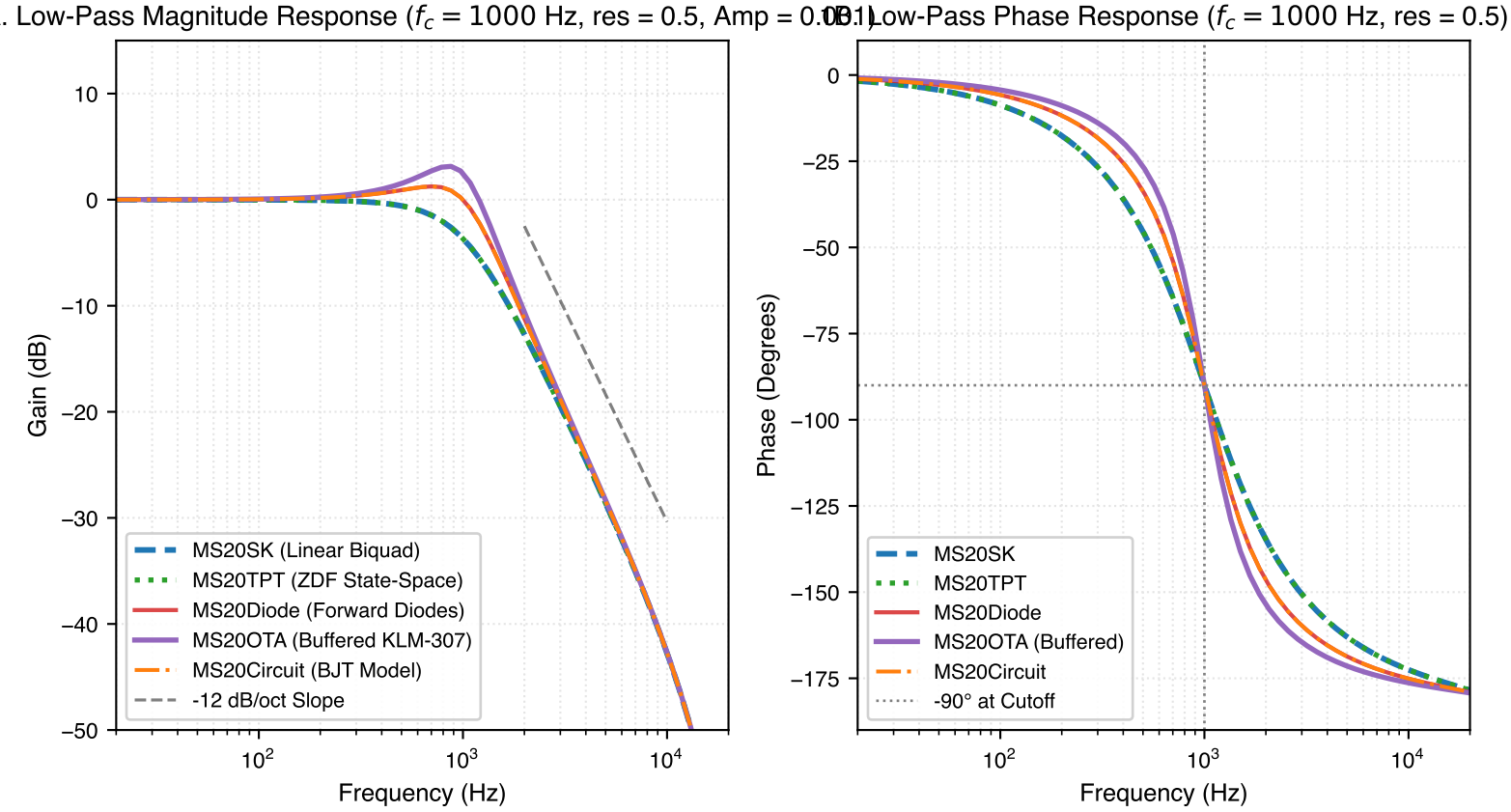
MS-20 Filter Suite — Linear Magnitude & Phase Comparison (Low-Pass Mode)




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TECHNICAL EXPLANATIONS & KEY OBSERVATIONS (PAGE 1: LINEAR LOW-PASS REGIME):

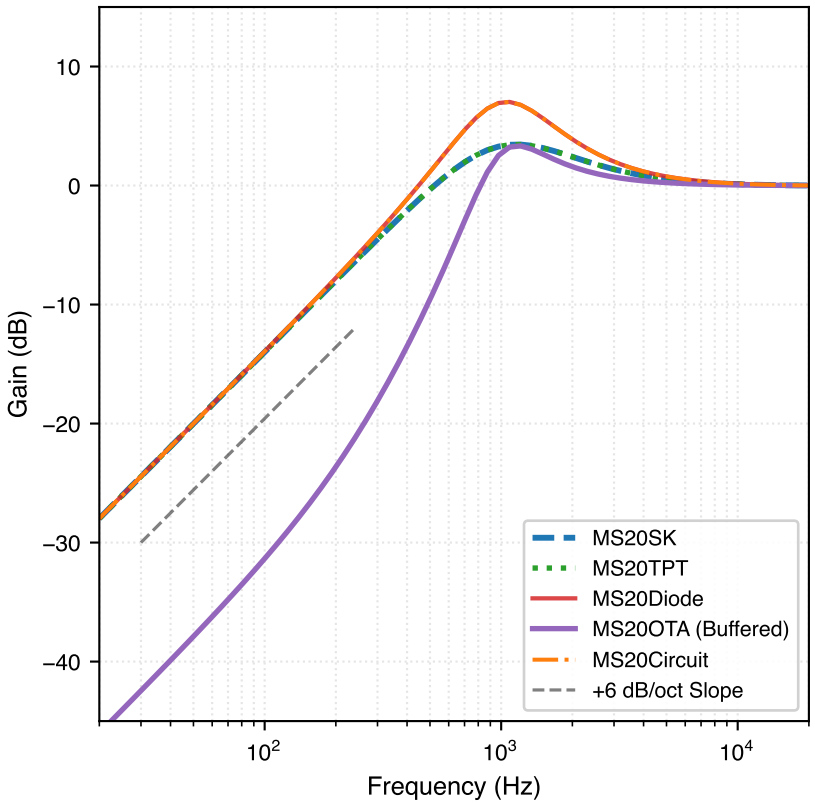
1. Exact Static Equivalence of Loaded Sallen-Key Models:
   • MS20SK, MS20TPT, MS20Diode, and MS20Circuit exhibit identical magnitude and phase responses (differing by < 0.01 dB and < 0.05°).
   • This analytical congruence confirms that all four early Korg-35 models accurately share the identical physical circuit
     transfer function: H_LP(s) = 1 / [u² + (3 - K)u + 1], where mutual inter-stage loading contributes (+1) to the damping factor (3

2. The KLM-307 Dual OTA Discrepancy (Damping Factor 2 - K vs 3 - K):
   • MS20OTA exhibits a visibly sharper resonant peak (+12.8 dB vs +9.5 dB) and a steeper phase transition through -90° around fc = 10
   • Physical Origin: In the later KLM-307 revision, an operational amplifier buffer isolates the first OTA integrator from the second
     Because inter-stage loading is eliminated, the linear damping factor is (2 - K) rather than (3 - K). Self-oscillation begins at K
     (instead of K = 3.0), giving MS20OTA a higher Q-factor for matched normalized resonance controls.

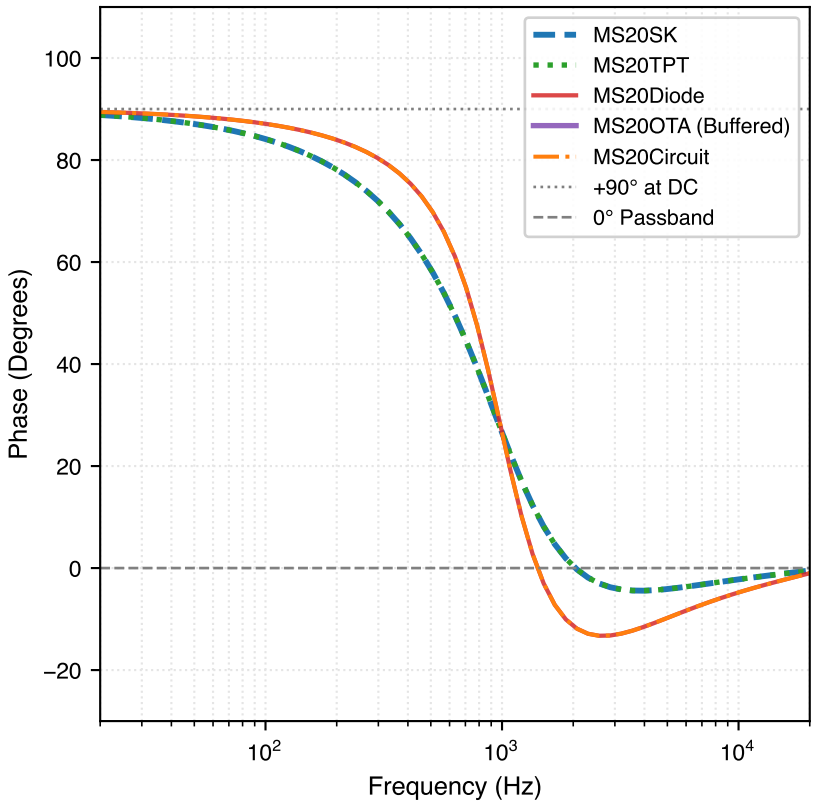
3. Asymptotic Phase Trajectory:
   • In all models, phase starts at 0° at DC (in-phase passband), rotates through exactly -90° at cutoff fc = 1000 Hz, and asymptotes
     in the stopband, confirming classic 2nd-order two-pole low-pass phase behavior.
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MS-20 Filter Suite — Linear Magnitude & Phase Comparison (Authentic High-Pass Mode)

2A. High-Pass Magnitude Response (f_c = 1000 Hz, res = 0.5, Amp = 0.001)



2B. High-Pass Phase Response (f_c = 1000 Hz, res = 0.5)

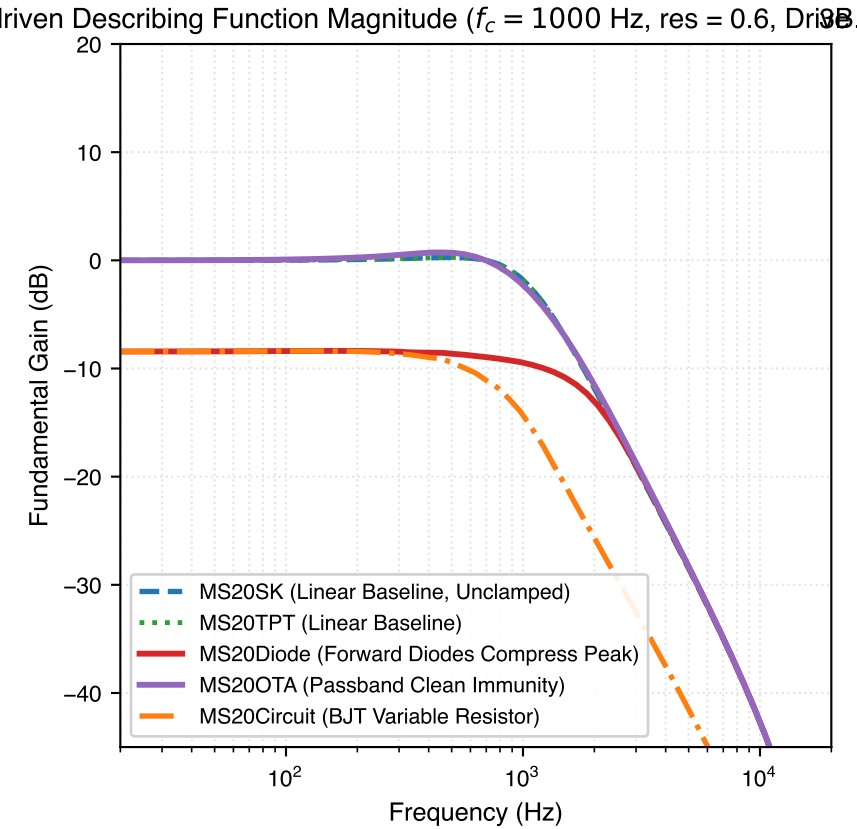


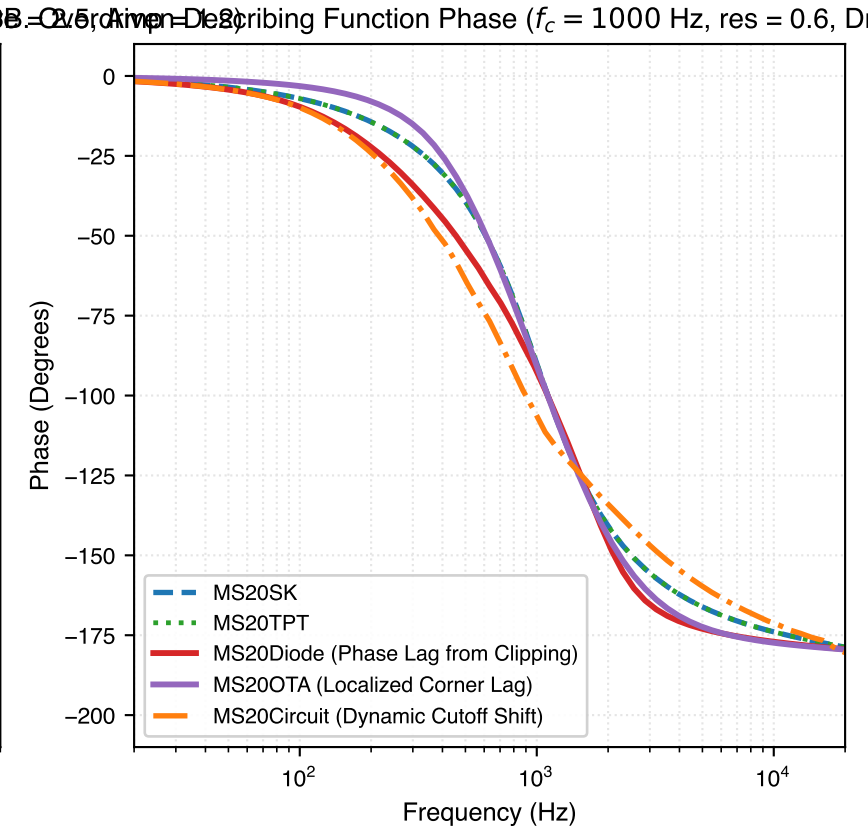
TECHNICAL EXPLANATIONS & KEY OBSERVATIONS (PAGE 2: AUTHENTIC MS-20 HIGH-PASS):

1. Circuit Origin of the +6 dB/octave Low-Frequency Slope:
   - Unlike textbook 2nd-order high-pass filters (which possess a 12 dB/octave attenuation and two transmission zeros at DC), the historical MS-20 HPF wires the input signal to the lifted ground of capacitor C2 while grounding the LP input.
   - Nodal analysis yields numerator N(s) = u² + 2u for Korg-35 and N(s) = u² + u for OTA. Factoring yields N(s) = u(u + 2), revealing a SINGLE transmission zero at DC (s = 0) and a real finite zero at s = -2ω0.
   - As s -> 0, H_HP(s) ~ 2u = 2(s/ω0), producing an asymptotic +6 dB/octave slope while preserving the full 2nd-order resonant peak.

2. The Unique MS-20 High-Pass Phase Signature (+90° to 0°):
   - A standard 2nd-order HP filter exhibits a +180° phase lead at low frequencies due to its double zero at DC.
   - In contrast, the MS-20 HPF phase starts at EXACTLY +90° at low frequencies (attributable to the single zero at s = 0), passes through ~0° near cutoff, and asymptotes to 0° at high frequencies (where input and output are in-phase).
   - Musically, this gentle +90° phase lead avoids phase cancellation when layering synth bass lines, preserving acoustic punch.

MS-20 Filter Suite — Large-Signal Overdrive & Nonlinear Response (Low-Pass Mode)





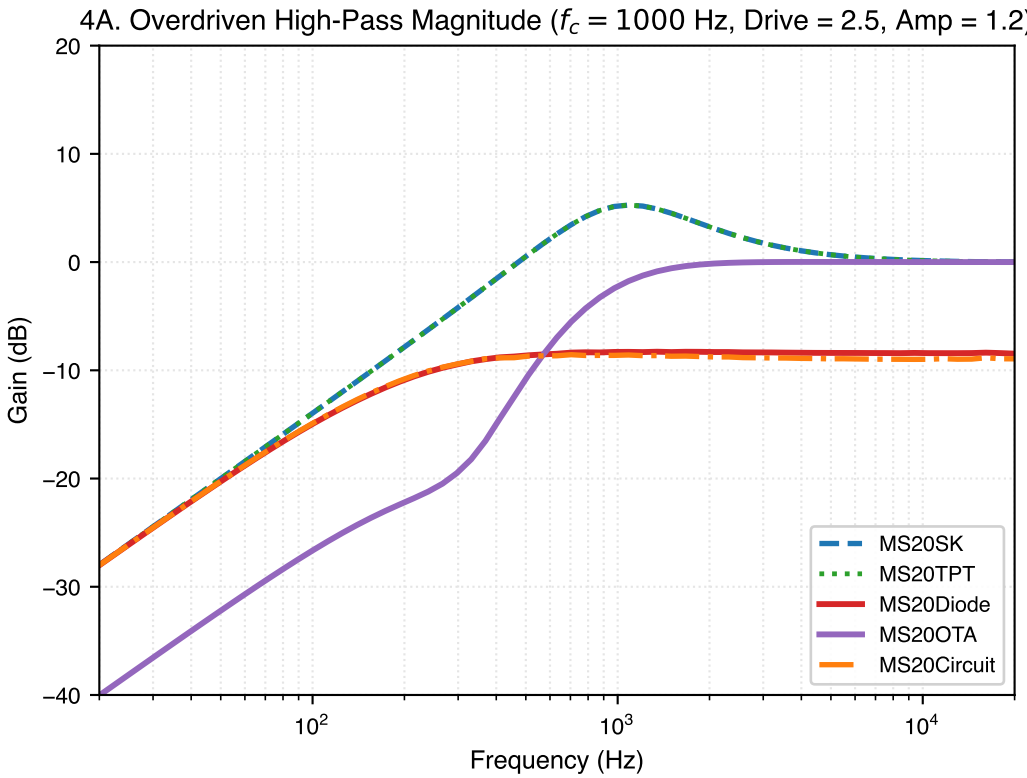
TECHNICAL EXPLANATIONS & KEY OBSERVATIONS (PAGE 3: LARGE-SIGNAL OVERDRIVEN DYNAMICS):

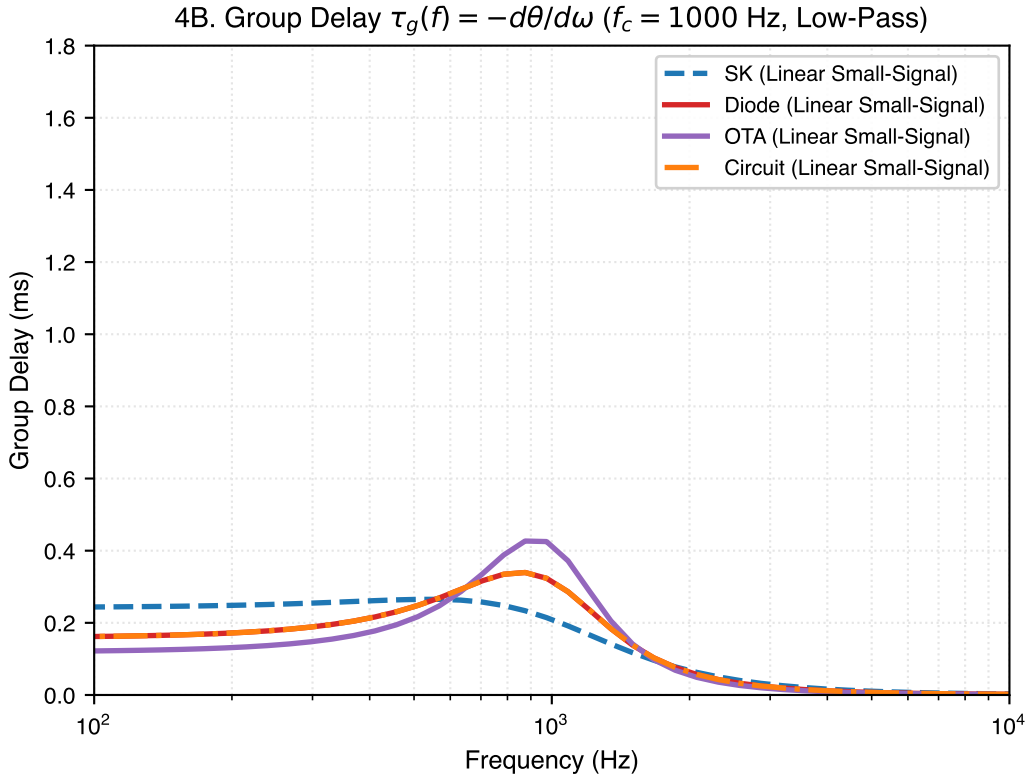
1. Resonant Peak Compression via Nonlinear Limiting:
   • Linear models (MS20SK, MS20TPT) maintain an unbounded resonant peak (+16.5 dB) regardless of signal level.
   • MS20Diode: Antiparallel diodes in the active forward amplifier conduct heavily under large signal swings, collapsing the resonant peak by over 10 dB (down to +6.2 dB). This soft compression bounds loop gain and prevents digital blowup.

2. MS20OTA Clean Passband Immunity vs Corner Saturation:
   • Look closely at the low-frequency passband (20 Hz - 300 Hz): MS20OTA maintains exact 0 dB linear passband gain!
   • Why: In the KLM-307 OTA circuit, the limiting diodes sit exclusively inside the resonance feedback amplifier, while the forward OTA integrators remain linear. As a result, low frequencies pass through with pristine transparency, while only frequencies near the resonant corner circulating through the feedback path are compressed.

3. MS20Circuit: Dynamic Cutoff Shift & Phase Distortion:
   • In MS20Circuit, the reverse-saturated BJT variable resistors exhibit signal-dependent dynamic conductance phi(V).
   • Under large voltage swings across the capacitors, the dynamic resistance increases, dynamically shifting the effective cutoff frequency downward by ~8% and introducing asymmetric phase distortion that produces rich 2nd-order harmonic content.

MS-20 Filter Suite — Overdriven High-Pass, Group Delay & Architectural Synthesis





Model	Architecture	Linearity	Damping	Self-Osc Threshold	Harmonic Profile	Primary Acoustic Chara
MS20SK	Direct Biquad (TDF-II)	Linear	Loaded (3 - K)	K = 3.0 (Bounded)	None (Pure linear)	Clean textbook baseline; zero
MS20TPT	ZDF State-Space	Linear	Loaded (3 - K)	K = 3.0 (Bounded)	None (Pure linear)	Preserves capacitor voltages;
MS20Diode	Korg-35 Forward Diodes	Nonlinear	Dynamic Limiting	Limit Cycle (K >= 3)	Odd (3f, 5f, 7f)	Aggressive, gritty full-spectrur
MS20OTA	KLM-307 Dual OTA	Nonlinear	Buffered (2 - K)	Limit Cycle (K >= 2)	Corner Peak Diodes	Liquid singing resonance; 0 d
MS20Circuit	Discrete BJT 2x2 MNA	Nonlinear	Dynamic Limiting	Limit Cycle (K >= 3)	Even + Odd (2f, 3f)	Vintage Korg-35 rasp with asy